

Anticipating adversaries in voting

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Outline

- 1 Introduction to voting
- 2 Formal set-up
- 3 Results
- 4 Appendix

A single winner election

- A set of voters $N = \{1, \dots, n\}$
- A set of candidates $X = \{\text{🇦🇩}, \text{👤}, \text{🍌}, \text{💀}\}$
- A type of ballots B : Ordinal ($\text{💀} \succ \text{🇦🇩} \succ \text{👤}$), Cardinal ($v_1(\text{🍌}) = 5, \dots$), Approval ($A_2 = \{\text{💀}, \text{🍌}\}$).
- A voting rule $f : B^N \rightarrow X$.

Plurality

Tops only, max number of supporters

voter	ballot				
1					
2					
3					
4					

Winner: 

Copeland

Count the number of won duels.

voter	ballot			
1		$\succ$		$\succ$ 
2		$\succ$		$\succ$ 
3		$\succ$		$\succ$ 
4		$\succ$		$\succ$ 

 is the Copeland winner.

Approval

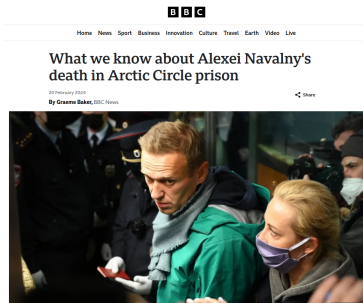
Select candidate that is approved by max number of voters

voter	ballot
1	
2	 ,  , 
3	 , 
4	 ,  , 

Winner is .

Positive Control

Adding/ Deleting candidates/voters for making some candidate win



voter	ballot			
1				
2				
3				
4				

voter	ballot			
3				
4				

Bribery



Faliszewski and Rothe [2016]

voter	ballot		
1			
2			
3			
4			

voter	ballot		
1			
2			
3			
4			

Types of Manipulation

Positive Control

- Target candidate $x \in X$
- Adding voters/candidates: set S of unregistered voters/candidates. Can we find $S' \subseteq S$ s.t. x wins ?
- Deleting voters/candidates

Bribery

- Target candidate $x \in X$
- Budget b
- Price system $(\pi_1, \dots, \pi_n)$
- $\pi_i : B \rightarrow \mathbb{R}$

Bribery price functions

A price function is defined on B which can be of exponential size ($m!$ orderings)

Different families of price functions

- discrete $\pi_i(\succ_i) = 0, \pi_i(\succ) = c_i$
- depends on the swaps (given value for pairs of candidates)

Example

 $\succ_i$
 $\succ_i$
 $\succ_i$
 $\neq$
 $\succ$
 $\succ$
 $\succ$


Suppose unit cost for each swap, then cost of 3.

Range of possibilities

One problem: for this election with selected candidates $X = \{\text{👤}, \text{🏠}, \text{👤}\}$, is it possible to make 🏠 a Plurality winner by adding dummy candidates in $S = \{\text{👤}, \text{🏠}\}$?

voter	ballot								
1									
2									
3									
4									

Immunity: Impossible to manipulate

Resistance: Manipulation problem is NP-hard

Vulnerability: Manipulation problem is in P

Positive Control

Theorem

<i>Type</i>	<i>A candidates</i>	<i>D candidates</i>	<i>A/D voters</i>
<i>Condorcet</i>	<i>Imumne</i>	<i>Vulnerable</i>	<i>Resistant</i>
<i>Approval</i>	<i>Imumne</i>	<i>Vulnerable</i>	<i>Resistant</i>
<i>Plurality</i>	<i>Resistant</i>	<i>Resistant</i>	<i>Vulnerable</i>

NB: Resistance results are proved from Exact Cover by 3 Sets

Bribery

Theorem

<i>Rule</i>	<i>Strength</i>
<i>Condorcet</i>	<i>Resistant</i>
<i>Approval</i>	<i>Resistant</i>
<i>Plurality</i>	<i>Vulnerable</i>

Good note: NP hard for weighted elections

Discussion

Is the complexity shield efficient ? No

What are the most common types of attack ?



What about strategy proofness ?

Good aspects of manipulation (prediction, campaign management)

References

P. Faliszewski and J. Rothe. Control and bribery in voting., 2016.

Appendix 1

Table 7.3. *The complexity of control problems for various voting rules*

Voting Rule	CAUC		CAC		CDC		CPC-TE		CPC-TP		CRPC-TE		CRPC-TP		CAV		CDV		CPV-TE		CPV-TP	
	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D
plurality	R	R	R	R	R	R	R	R	R	R	R	R	R	R	V	V	V	V	V	V	R	R
(Bartholdi et al., 1992; Hemaspaandra et al., 2007)																						
Condorcet	I	V	I	V	V	I	V	I	V	I	V	I	V	I	R	V	R	V	R	V	R	V
(Bartholdi et al., 1992; Hemaspaandra et al., 2007)																						
approval	I	V	I	V	V	I	V	I	I	I	V	I	I	I	R	V	R	V	R	V	R	V
(Hemaspaandra et al., 2007)																						
Copeland ^a																						
for $\alpha = 0$	V	V	R	V	R	V	R	V	R	V	R	V	R	V	R	R	R	R	R	R	R	R
$0 < \alpha < 1$	R	V	R	V	R	V	R	V	R	V	R	V	R	V	R	R	R	R	R	R	R	R
$\alpha = 1$	V	V	R	V	R	V	R	V	R	V	R	V	R	V	R	R	R	R	R	R	R	R
(Faliszewski et al., 2009c)																						
maximin	V	V	R	V	V	V	-	-	-	-	-	-	-	-	R	R	R	R	-	-	-	-
(Faliszewski et al., 2011b)																						

Appendix 2

Table 7.3. *The complexity of control problems for various voting rules*

Voting Rule	CAUC		CAC		CDC		CPC-TE		CPC-TP		CRPC-TE		CRPC-TP		CAV		CDV		CPV-TE		CPV-TP	
	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D
Borda	–	–	R	V	R	V	–	–	–	–	–	–	–	–	R	V	–	V	–	V	–	–
(Russel, 2007; Elkind et al., 2011a; Loreggia et al., 2014; Chen et al., 2015)																						
SP-AV	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	V	R	V	R	V	R	R
(Erdélyi et al., 2009)																						
fallback	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	V	R	V	R	R	R	R
(Erdélyi and Rothe, 2010; Erdélyi et al., 2011; see also Erdélyi et al., 2015a)																						
Bucklin	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	V	R	V	R	R	R	S
(Erdélyi et al., 2011; see also Erdélyi et al., 2015a)																						
RV	I	V	I	V	V	I	V	I	I	I	V	I	I	I	R	V	R	V	R	V	R	V
(Menton, 2013)																						
NRV	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	V	R	V	R	R	R	R
(Menton, 2013)																						
Schulze	R	S	R	S	R	S	R	V	R	V	R	V	R	V	R	V	R	V	R	R	R	R
(Parkes and Xia, 2012; Menton and Singh, 2013)																						

Appendix 3

Table 7.4. *The complexity of f -BRIBERY for various voting rules*

f	f -BRIBERY	reference
plurality	P	Faliszewski et al. (2009b)
veto	P	Faliszewski et al. (2009b)
2-approval	P	Lin (2012)
k -veto, $k \in \{2, 3\}$	P	Lin (2012)
k -approval, $k \geq 3$	NP-complete	Lin (2012)
k -veto, $k \geq 4$	NP-complete	Lin (2012)
Borda	NP-complete	Brelsford et al. (2008)
STV	NP-complete	Xia (2012a)
Bucklin	NP-complete	Faliszewski et al. (2015)
fallback	NP-complete	Faliszewski et al. (2015)
maximin	NP-complete	Faliszewski et al. (2011b)
Copeland	NP-complete	Faliszewski et al. (2009c)
Schulze	NP-complete	Parkes and Xia (2012)
ranked pairs	NP-complete	Xia (2012a)
approval	NP-complete	Faliszewski et al. (2009b)
range voting	NP-complete	follows from the approval result